
Scaling Empowered Agents

Catherine Ji*

Department of Physics
Princeton University
cj7280@princeton.edu

Vivek Myers*

Department of Computer Science
UC Berkeley
vmyers@berkeley.edu

Sergey Levine

Department of Computer Science
UC Berkeley

Benjamin Eysenbach

Department of Computer Science
Princeton University

August 17, 2026

Abstract

Empowerment measures the capacity of an agent to actively control its environment. This object has been identified as an unsupervised mechanism for natural and artificial agents to make sense of large environments. In this work, we scale empowerment maximization to large discrete environments and high-dimensional, continuous environments. In addition to exhibiting exploration behavior, empowered agents seek out states in the environment that grant high degrees of control. Our results also show that empowerment depends strongly on the underlying state representation and should not be arbitrarily maximized over full state spaces. Our work supports empowerment maximization as a principled mechanism to train unsupervised agents in large environments.

1 Introduction

In large state spaces, searching for sparse rewards is a hard, often impossible task. An undirected agent often cannot stumble upon interesting behavior using random actions. However, real environments have inherent structure that makes the problem of searching for sparse reward more tractable. For instance, an agent may wish to accumulate resources and gain control over certain objects in an environment, so these objects may serve as a resource for arbitrary future task; most interesting tasks in an environment (e.g. via externally-defined rewards) often align with the same degrees of freedom as resource accumulation. In this work, we introduce a method to train agents that maximizes the generic, universal resource of *control*: we reward an agent for its ability to exert control over an environment. We quantify control using the information-theoretic quantity of **empowerment**, and build on state-of-the-art skill learning methods to maximize empowerment in large environments.

Empowerment measures the degree to which an agent can control their environment. Prior literature has shown that empowerment maximization can help explain how natural and artificial agents build causal models of the world and interact with large environments [1?, 2]. Indeed, empowerment speaks to a general need for RL methods that are compatible with the Big World Hypothesis [3]: real-world agents should successfully interact with an environment that they cannot fully describe. Despite a long line of literature studying the behavior of empowerment-maximizing agents, there has not yet been empirical demonstrations of empowerment maximization in such challenging, large settings that escape simple characterization. Early literature (VIC [4] and its predecessor) estimate and maximize empowerment

*Equal contribution.

as a *channel capacity* using variational methods [5–7]. However, most of these works only scale up to simple tabular settings or low dimensional continuous settings. These methods eventually gave rise to a later line of unsupervised skill discovery literature successful in higher-dimensional locomotion and manipulation tasks [8? –10].

We aim to close this gap between concept and current practice. Our starting observation is that state-of-the-art mutual information skill learning methods already maximize quantities similar to empowerment; MISL maximizes value functions that lower bound empowerment. In this work, we present a recipe to adapt unsupervised skill learning methods to maximize empowerment in challenging, high-dimensional settings. The goal of this work is to provide methods to tractably train empowerment-maximizing agents. Using modern skill learning methods, we present strong exploration results in Craftax-Classic [11]. Furthermore, our empirical results show that empowerment maximization rewards behavior distinct from canonical intrinsic exploratory methods, prioritizing resource and skill accumulation over novelty.

Our work suggests that empowerment maximization is a tractable mechanism for agents to explore large environments. We present future directions to make empowerment maximization methods more robust and scalable, particularly in the continuous setting.

2 Related Work

Intrinsically-Motivated Exploration Methods. Such methods are classically used in settings with sparse or absent rewards, where intrinsic signals can help synthesize complex behaviors and can be roughly categorized into curiosity, novelty, information gain, and prediction error [12]. Intrinsically motivated methods are exploration methods in reinforcement learning (RL) that require no external information about the Markov decision process (MDP), outside of environment interaction [12]. Such methods are classically used in settings with sparse or absent rewards, where intrinsic signals can help synthesize complex behaviors and can be roughly categorized into curiosity [13], novelty [14], information gain [15], and prediction error [16]. While empowerment has also been proposed as an intrinsic motivation [5], there remain fundamental challenges scaling this intrinsic reward beyond tabular and simple continuous settings. Such methods are classically used in settings with sparse or absent rewards, where intrinsic signals can help synthesize complex behaviors and can be roughly categorized into curiosity [13], novelty [14], information gain [15], and prediction error [16]. While empowerment has also been proposed as an intrinsic motivation [5], there remain fundamental challenges scaling this intrinsic reward beyond tabular and simple continuous settings.

Unsupervised Skill Discovery. Unsupervised skill discovery methods learn latent-conditioned policies without access to extrinsic (supervised) reward signals. Mutual Information Skill Learning (mutual information skill learning (MISL)) forms a subset of these methods, learning policies that maximize mutual information (MI) between observations and actions [8]. Generally, MISL methods learn policies such that learned latents are informative of future states.

Potential and Effective Empowerment. Classically, empowerment quantifies the influence an agent’s actions can exert on the future state distribution of the environment [6?]. This is formalized as a channel capacity (maximal directed information over actions) between the sequence of actions and future states [17]. As originally defined, potential empowerment is the channel capacity at a particular state, quantifying how much an agent can control from that state.

In works from psychology and cognitive sciences, empowerment has been conjectured to explain human behavior in complex, open-ended settings not captured by other intrinsic motivations like curiosity [1, 18–20]. For example, an agent seeking to control its environment would not waste effort staring at random noise (the Noisy TV problem), a desirable property for both human-like and assistance agents [21, 22]. Several works have proposed empowerment and related objectives as suitable for

training artificial agents. Prior work on estimating and maximizing empowerment includes variational approaches using general logit parameterizations like VIC [4], though these methods do not scale to high-dimensional continuous settings. These original works on empowerment maximization gave rise to line of unsupervised skill discovery work now known as Mutual Information Skill Learning, or MISL, methods [8–10].

Skill Learning and Effective Empowerment. MISL objectives have a close relationship to empowerment. Importantly, the effective empowerment objective and skill learning objectives are similar, even though modern skill learning methods do not explicitly reference empowerment. Indeed, predecessors to modern MISL methods including VIC [4, 5] explicitly aim to maximize forms of empowerment. Modern MISL methods optimize effective empowerment by learning skills that actively transmit information to the future.

Effective empowerment lower bounds potential empowerment [23]. However, prior empowerment maximization approaches did not scale to high-dimensional continuous settings. In this work, we aim to close this gap by optimizing empowerment using modern MISL methods in large, open-ended settings—the setting where empowerment matters. We introduce methods and provide both on-policy and off-policy implementations, highlighting key practical considerations: (1) empowerment over observation versus state space, (2) defining empowerment in continuous action spaces, and (3) breaking symmetry in empowerment-driven policies.

The key contribution of our paper is adapting modern MISL approaches to empowerment maximization, demonstrating that empowerment maximization is tractable in large, open-ended settings. Our work opens empowerment maximization as a practical objective and advances research on intrinsically-motivated agents in open-ended environments.

3 Preliminaries

MDP problem setting. We present notation for a fully observed MDP with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, and environment dynamics $p(s' | s, a)$.

The latent skill z lies in a continuous skill space $\mathcal{Z} \subseteq \mathbb{R}^d$, taken to be the unit hypersphere in our continuous-control experiments. We fix $z \sim p(z) = \text{Unif}(\mathcal{Z})$ and write $\Pi_{\mathcal{Z}}$ for the class of skill- and start-state-conditioned policies $\pi(a | s, z; s_0)$, where s_0 is the context of the skill [24]. We write Π for policies $\pi(a | s)$ and fix a discount factor $\gamma \in (0, 1)$.

Shifted discounted state occupancy. For an initial state s_0 , skill $z \in \mathcal{Z}$, and policy $\pi(a | s, z; s_0)$, define the discounted state occupancy measure (DSOM)

$$p^\pi(s_+ = s | s_0, z) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p^\pi(s_{t+1} = s | s_0, z). \quad (1)$$

Averaging over a fixed skill source gives

$$p^\pi(s_+ | s_0) = \mathbb{E}_{p(z)} [p^\pi(s_+ | s_0, z)]. \quad (2)$$

Potential and effective empowerment. We distinguish potential empowerment, the maximum achievable MI from a state, from effective empowerment, the MI realized by a fixed skill-conditioned policy [17, 25].

Definition 3.1 (Potential empowerment). Let $Z \sim p(z) = \text{Unif}(\mathcal{Z})$ and let $S_+ \sim p^\pi(s_+ | s_0, Z)$. Potential empowerment is

$$\mathcal{E}_{\text{pot}}(s_0) \triangleq \max_{\pi \in \Pi_{\mathcal{Z}}} I^\pi(Z; S_+ | S_0 = s_0). \quad (3)$$

For continuous $\mathcal{Z}$, fixing the uniform prior loses no generality. However, our experiments use finite set of skills for simplicity. This is a practical assumption that does lose generality, instead approximating channel capacity on a space of discretized source distributions.

Potential-policy value functions. For any policy $\pi \in \Pi$, define

$$Q_{\text{pot}}^{\pi}(s, a) \triangleq \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{E}_{\text{pot}}(s_{t+k}) \mid s_t = s, a_t = a \right], \quad (4)$$

$$V_{\text{pot}}^{\pi}(s) \triangleq \mathbb{E}_{a \sim \pi(a|s)} [Q_{\text{pot}}^{\pi}(s, a)]. \quad (5)$$

The definition of a potential empowerment maximizing policy follows:

Definition 3.2 (Potential empowerment policy). A potential empowerment policy $\pi_{\text{pot}} \in \Pi$ takes the potential empowerment of Definition 3.1 as a reward.

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} V_{\text{pot}}^{\pi}(s) \quad \text{for all } s \in \mathcal{S}. \quad (6)$$

While conceptually similar, mutual information skill learning methods do not maximize potential empowerment, and maximize effective empowerment. *Effective empowerment* captures the MI itself:

Definition 3.3 (Effective empowerment). For a fixed policy $\pi \in \Pi_{\mathcal{Z}}$, effective empowerment at s_0 is the MI realized under the same uniform skill source:

$$\mathcal{E}_{\text{eff}}(\pi, s_0) \triangleq I^{\pi}(Z; S_+ \mid S_0 = s_0) \quad (7)$$

$$= \mathbb{E}_{p(z)} D_{\text{KL}} [p^{\pi}(s_+ \mid s_0, z) \parallel p^{\pi}(s_+ \mid s_0)]. \quad (8)$$

Effective empowerment contains no maximization and measures the MI achieved by a particular policy. Consequently,

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} \mathcal{E}_{\text{eff}}(\pi, s_0). \quad (9)$$

4 Methods

MISL methods learn skill-conditioned policies π_{eff} and value function that measure the *effective empowerment* from a state s_0 (Definition 3.3). These skills therefore serve as the yardsticks of empowerment: the diversity of these skills provides a lower bound on the potential empowerment of s_0 . A *potential empowerment* policy π_{pot} instead visits states with high potential empowerment:

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}(s_t) \right]. \quad (10)$$

This connection suggests a two-policy algorithm. The skill policy π_{eff} estimates a lower bound $\widehat{\mathcal{E}}_{\text{pot}}(s)$ on statewise potential empowerment, while the potential policy π_{pot} maximizes this estimate as an intrinsic reward. The effective and potential empowerment policies therefore support one another: the skills π_{eff} measure the controllable futures available from a state, while the potential empowerment policy π_{pot} moves the agent toward states from which the agent can exert control.

Algorithm 1: Empowerment maximization. Skills estimate the degree of control from each starting state. The potential policy π_{pot} ascends this estimate. Full version in Algorithm 2.

-
- 1: **for** $n = 1, \dots, N$ **do**
 - 2: **Skills:** train π_{eff} using $Q_{\text{eff},z}^{\pi_{\text{eff}}}$
 - 3: **Estimate:** $\widehat{\mathcal{E}}_{\text{pot}}(s) \leftarrow \mathbb{E}_{p(z)} [V_{\text{eff},z}^{\pi_{\text{eff}}}(s; s)]$
 - 4: **Ascend:** train π_{pot} using $\widehat{Q}_{\text{pot}}^{\pi_{\text{pot}}}$
-

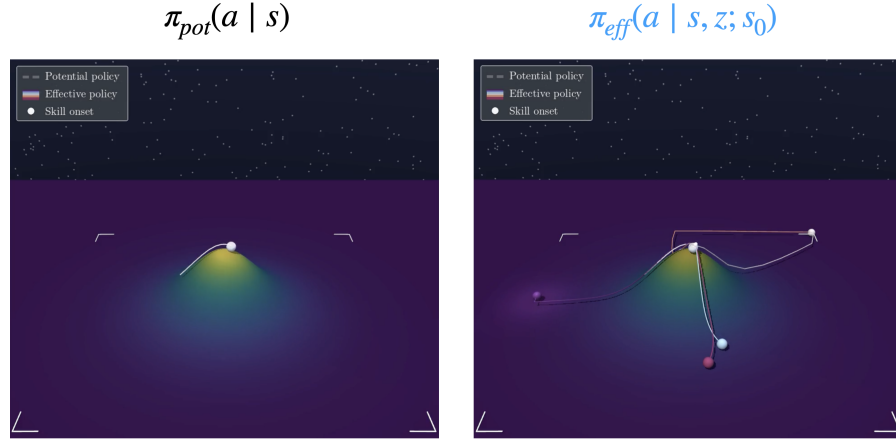


Figure 1: **Two trained empowerment-maximizing policies characterize central states in an MDP.** Consider the MDP of an agent (ball) on a hill, where the top represents a clearing. (Left) The potential empowerment policy (white) ascends the empowerment landscape, using the per-state empowerment $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (Right) The effective empowerment policy π_{eff} (colors) maximizes $I^\pi(Z; S_+ | S_0)$ over skills z , and provides a lower bound on the potential empowerment.

High-level method summary. The empowerment-maximization algorithm (Algorithm 1) alternates between training the two policies shown in Fig. 1. First, we train the skills π_{eff} and use their on-policy values to estimate a lower bound $\widehat{\mathcal{E}_{\text{pot}}}(s)$ on potential empowerment. We then use this estimate as an intrinsic reward for the potential policy π_{pot} . For our chosen MISL objective, the skill update is an effective-empowerment policy-improvement step under the assumptions of Theorem B.1, a modified statement of the policy improvement results in [4]. Thus, MISL methods can lead to policy improvement on a lower bound of the potential empowerment.

Training structure. However, while we have a general idea of the objects to optimize, key questions remain in building a scalable implementation of empowerment maximization:

1. How do we gather data to train the empowerment objective?
2. What are practical implementation details necessary to get empowerment maximization methods to work in high DoF settings?

Switching policies. While we presented a methodology to convert skill-learning methods into potential empowerment maximization algorithms, we cannot simply learn skills for every starting state s_0 , bound the potential empowerment using the effective empowerment, then maximize the potential empowerment. For arbitrarily large state spaces, learning skills everywhere is exactly the type of intractable exploration problem that empowerment maximization wishes to avoid.

We consider a two-stage rollout scheme where we only consider effective empowerment maximization at reachable states. During data collection, we roll out both policies within a single trajectory. The potential policy first moves the agent to a handoff state s_0 , after which the agent switches “on” the skills by sampling a skill z and executing $\pi_{\text{eff}}(\cdot | \cdot, z; s_0)$ (Fig. 1). The two phases serve distinct roles. The potential policy determines *where* the agent initiates skills, while the skill policy measures the controllable futures available from that state. Both of these behaviors are important. The potential policy enables the agent to modulate the starting state for the skills and estimate the empowerment *field*. Meanwhile, skills lower bound the empowerment of any fixed state. We can interpret skills as local probes of empowerment, where the potential empowerment policy ascends the empowerment landscape given by these probes. The two-phase rollout naturally constrains the skill learning to the reachable states and provides a useful training signal for the potential policy in large environments.

4.1 Mutual Information Skill Learning

Contrastive Successor Features. We consider variants of Zheng et al. [10, contrastive successor features (CSF)], a recent state-of-the-art skill-learning method, to lower-bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ \mid S_0 = s_0)$ for a fixed starting state s_0 . CSF estimates the one-step MI $I(Z; S, S')$ from a contrastively-trained log density ratio; we instead train the classifier on transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$, which gives the long-horizon MI conditioned on the handoff state. Fixing the skill prior to $p(z \mid s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ removes the otherwise intractable $p(z \mid s_0)$ from the estimator (Definition 3.1). The full derivation is in §. A.3.

Following CSF [10] and METRA [9], we parametrize the classifier logits with a bilinear form in terms of representations ϕ :

$$f_\phi(z; s_0, s_+) = z^\top (\phi(s_+) - \phi(s_0)) \approx \log \frac{q_\phi(z \mid s_0, s_+)}{p(z)}. \quad (11)$$

This parameterization forces skills to spread across linearly-separable directions in representation space; the geometric benefits of such a parameterization are discussed in [9]. We train f_ϕ with InfoNCE [26] when $\mathcal{Z}$ is continuous and multi-class cross-entropy when the skill space is finite. Importantly, the classifier may not receive the same set of information as the policy and critic: for instance, we may stipulate that only changes over certain relevant degrees of freedom can identify skills while policies and value functions take the whole state as input.

Effective-policy value functions. For a fixed classifier f_ϕ , define the normalized effective empowerment reward

$$r_{\text{eff},\phi}(s, a, s'; z, s_0) \triangleq (1 - \gamma) f_\phi(z; s_0, s'). \quad (12)$$

In other words, the reward measures the relative probability that a transition arose from the executed z (rather than the marginal z). The corresponding value functions are

$$Q_{\text{eff},z}^\pi(s, a; s_0) \triangleq \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{\text{eff},\phi}(s_{t+k}, a_{t+k}, s_{t+k+1}; z, s_0) \mid s_t = s, a_t = a \right], \quad (13)$$

$$V_{\text{eff},z}^\pi(s; s_0) \triangleq \mathbb{E}_{a \sim \pi(a \mid s, z; s_0)} [Q_{\text{eff},z}^\pi(s, a; s_0)]. \quad (14)$$

Training the skills. Averaging the start-state values over skills gives an immediate interpretation of these value functions. The value function is the variational lower bound on the effective empowerment

$$\mathbb{E}_{p(z)} [V_{\text{eff},z}^\pi(s_0; s_0)] \leq \mathcal{E}_{\text{eff}}(\pi, s_0). \quad (15)$$

Under the assumptions of Theorem B.1, alternating policy improvement with on-policy classifier fitting improves the true MI. This variational lower bound has been shown previously in [4].

Training the potential policy. By definition of potential empowerment, the effective empowerment of any policy from any starting state lower bounds the potential empowerment at that same state. Thus, a variational lower bound on effective empowerment lower bounds the potential empowerment:

$$\widehat{\mathcal{E}}_{\text{pot}}(s) \triangleq \mathbb{E}_{p(z)} [V_{\text{eff},z}^\pi(s; s)] \leq \mathcal{E}_{\text{eff}}(\pi_{\text{eff}}, s) \leq \mathcal{E}_{\text{pot}}(s). \quad (16)$$

The potential empowerment serves as the reward for the potential empowerment policy π_{pot} . We train π_{pot} using $\widehat{\mathcal{E}}_{\text{pot}}$ as its intrinsic reward, where $\widehat{Q}_{\text{pot}}^\pi$ and $\widehat{V}_{\text{pot}}^\pi$ are the corresponding value functions. In our experiments, we use SAC for the continuous-control settings [27] and PPO for the discrete settings [?].

In summary, we follow the general formula of using CSF, a scalable MISL method, to lower bound effective empowerment, or the mutual information realized by skill-conditioned policies. Because the effective empowerment, in turn, is a lower bound on potential empowerment, these skills serve as local probes of the potential empowerment. Then, we train a second potential empowerment policy that explicitly ascends this probed landscape, where this agent navigates to states *from which* it can exert control.

4.2 Empowerment in High-Dimensional Settings

With these policy and value function objects in hand, we address two important implementation details that enable the method to scale.

Hindsight relabeling. The effective empowerment policy $\pi_{\text{eff}}(a \mid s; z, s_0)$ conditions on both the skill and the handoff state. This space can be enormous. To reuse previously collected trajectories, we sample an earlier handoff state $\tilde{s}_0$ from the same trajectory and relabel the skill as $\tilde{z} \in \arg \max_z f_\phi(z; \tilde{s}_0, s_+)$. We update the effective policy and critic using the relabeled tuple, following the hindsight-relabeling principle of Andrychowicz et al. [28]. Unfortunately, hindsight relabeling introduces bias, and, in practice, overly concentrates skills, reducing diversity. Thus, we weight each relabeled sample by the inverse effective number of samples of the assigned skill following [29] which helps reduce skill collapse while improving sample efficiency.

Classifier bottleneck. In continuous state spaces, small perturbations can yield high MI without visually distinguishable behavior. We therefore add Gaussian noise to selected state coordinates before passing s_0 and s_+ to the classifier. This is effectively a simplified version of VIB [30] that regularizes the information from the state space to the representation considered in empowerment maximization.

5 Maximizing empowerment in high-dimensional, continuous settings

We evaluate the method in large discrete and continuous environments. We ask:

1. Can we tractably maximize empowerment in large, high-dimensional environments?
2. Do potential empowerment policies π_{pot} seek states that, intuitively, have high controllability? Do effective empowerment policies π_{eff} learn distinguishable skills?
3. How does empowerment depend on the chosen representation *over which* empowerment is defined?

We include training setup and hyperparameters in ??.

Scalability. We provide demonstrations of scaling the empowerment maximization method to high-dimensional, discrete settings. Craftax-Classic [?] is a large and partially observed setting with a 1345-dimensional symbolic observation and 17 discrete actions (Fig. 2a) consisting of crafting, resource collection, fighting mobs, and eating/drinking. Using only an empowerment reward defined over the *agent's inventory* and a discrete set of 16 skills, the agent unlocks on average 10.6 ± 0.1 out of the 22 total achievements in a single two-phase rollout (mean $\pm$ s.e.m. across 5 seeds, 10^9 environment steps). This intrinsic reward exceeds the best previously-reported success rates of intrinsic reward methods at the same environment-step budget [31]. The potential empowerment estimate also rises steadily over training to 1.9 ± 0.1 nats of the $\ln 16 \approx 2.8$ nat ceiling (Figs. 4g and 4h), tracking the development of interesting behavior. Implementation details and hyperparameters for Craftax-Classic are in §. C.

To test whether the method works in continuous settings, we chose two high-DoF benchmarks: AntSoccer [33] (Fig. 3b) and BuilderBench [32] (Fig. 2b). Both environments represent common RL benchmarks for locomotion and manipulation and lead to different types of empowering behaviors. In both settings, the empowerment policies learn visually distinguishable and interpretable behavior that spread over the state space (Figs. 2b and 3).

Behavioral interpretation. The AntSoccer environment consists of an ant and a ball. The policy π_{pot} learns to navigate to the *center of the space and stay there* (white, Fig. 3a) while the policies π_{eff} branch and cover the space (colors, Fig. 3a). Importantly, the behavior π_{eff} becomes more distinguishable because the agent ascends the potential empowerment landscape towards the center of the arena: the agent is more potentially empowered at the center and can cleanly execute many different skills. This result supports our intuition that potential and effective empowerment policies support each other, where

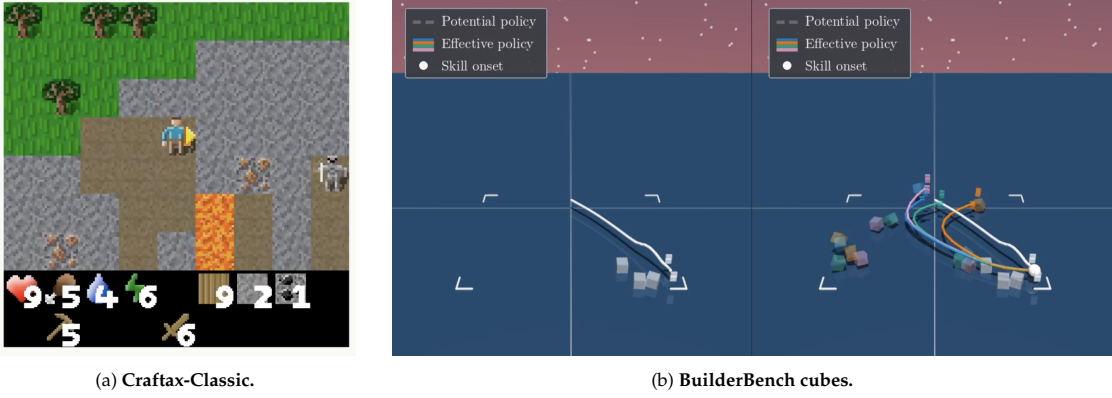


Figure 2: **Empowerment maximization in large discrete and continuous environments.** (a) Potential empowerment policy in Craftax-Classic [11]. The policy learns to gather resources, craft tools, then use tools to collect more resources. (b) Potential and effective empowerment policies in BuilderBench [32] with cubes in the corner. (Left) The potential policy navigates the gripper behind the cubes and stays there, preparing to control the cubes. (Right) The effective policies knock the cubes in different directions.

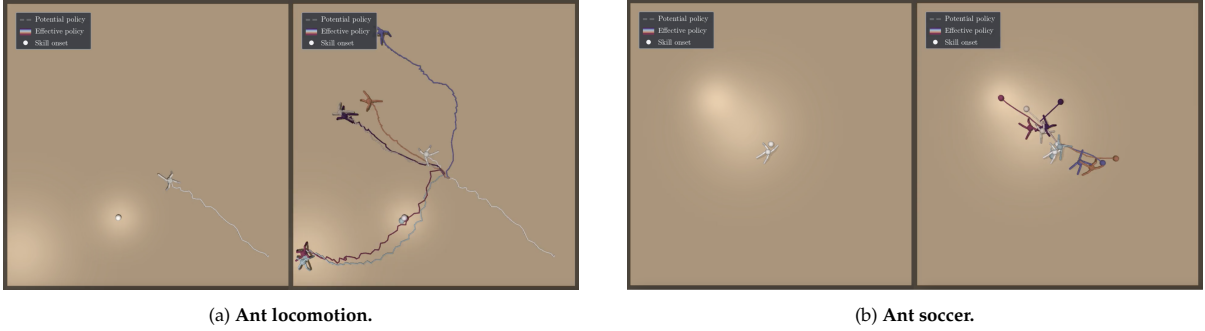


Figure 3: **Empowerment maximization on high-DoF Ant tasks** [33]. (a) Ant locomotion: skills cover diverse movement directions, and the potential policy navigates to states with max directional diversity. (b) Ant soccer: skills kick ball in different directions, and the potential policy positions the agent to kick and dribble the ball.

effective empowerment policies act as probes of potential empowerment. The average (lower bound on the) estimated potential MI grows over training time as the ant learns to walk, demonstrating that the policy is successfully maximizing the empowerment objectives (Figs. 4a and 4b).

BuilderBench [32] tests manipulation and consists of a gripper with cubes. The environment we study contains 3 cubes in the corner of a boxed play area (Fig. 2b, white corners denote the boxed play area). Defining the empowerment over the cube xyz positions leads to similar controllability-seeking behavior. The potential empowerment policy sneaks behind the blocks, where it has easy access to the blocks, and stays there. The effective empowerment policies then knock the blocks in different directions. The potential empowerment estimate increases over the course of training as the arm learns to interact with the blocks (Figs. 4e and 4f).

Empowerment depends on the representation. While the above results demonstrate that empowerment maximization can lead to interesting agent behavior, where agents learn to interact with objects and reach central states in an MDP, empowerment can exploit fine-grained perturbations in the chosen representation and reward trivial behavior. This becomes particularly problematic in continuous or large settings – e.g. skills can maximize MI while perturbing the environment in small, meaningless ways.

More generally, the representations that parameterize empowerment can drastically change the locations of potentially-empowered states. For example, consider an experiment restricting the representation underlying empowerment in AntSoccer. If the encoder must identify skills from changes in the position

of the soccer ball, the agent is only empowered if it learns to manipulate and control the ball. Rather than navigating to a physically central state within the arena, the potential empowerment policy now positions the ant to approach and kick the ball, which, after the agent learns to interact with the ball, becomes the new centralized state (Fig. 3b, left; §. B.4.2). This behavior is markedly different from the full-observation setting, where the learned centralized state is the typical center of the arena, where the agent can easily execute skills such as running in arbitrary directions. Conversely, in Craftax Classic, we consider a similar experiment that *extends* observations from the inventory to the entire observation space, which includes degrees of freedom such as agent orientation and position. While MI quickly approaches $2 \text{ nats} < \ln 16 \text{ nats}$, the agent no longer needs substantial resource accumulation or crafting to achieve high empowerment, since the agent is equally happy with simpler options including running in particular directions, controlling its rate of health depletion, and so on.

These experiments demonstrate that the usefulness of empowerment in large environments depends on the underlying representation. By itself, the empowerment objective does not distinguish between semantically meaningful and incidental controllable degrees of freedom. It is therefore natural to hypothesize that intelligent agents maximize empowerment over degrees of freedom *that align with a form of goal reaching or distribution of task rewards*. Thus, these experimental results suggest that a useful notion of empowerment should reward control over degrees of freedom selected by a separate representation learning process; *which* set of representations remains an open problem.

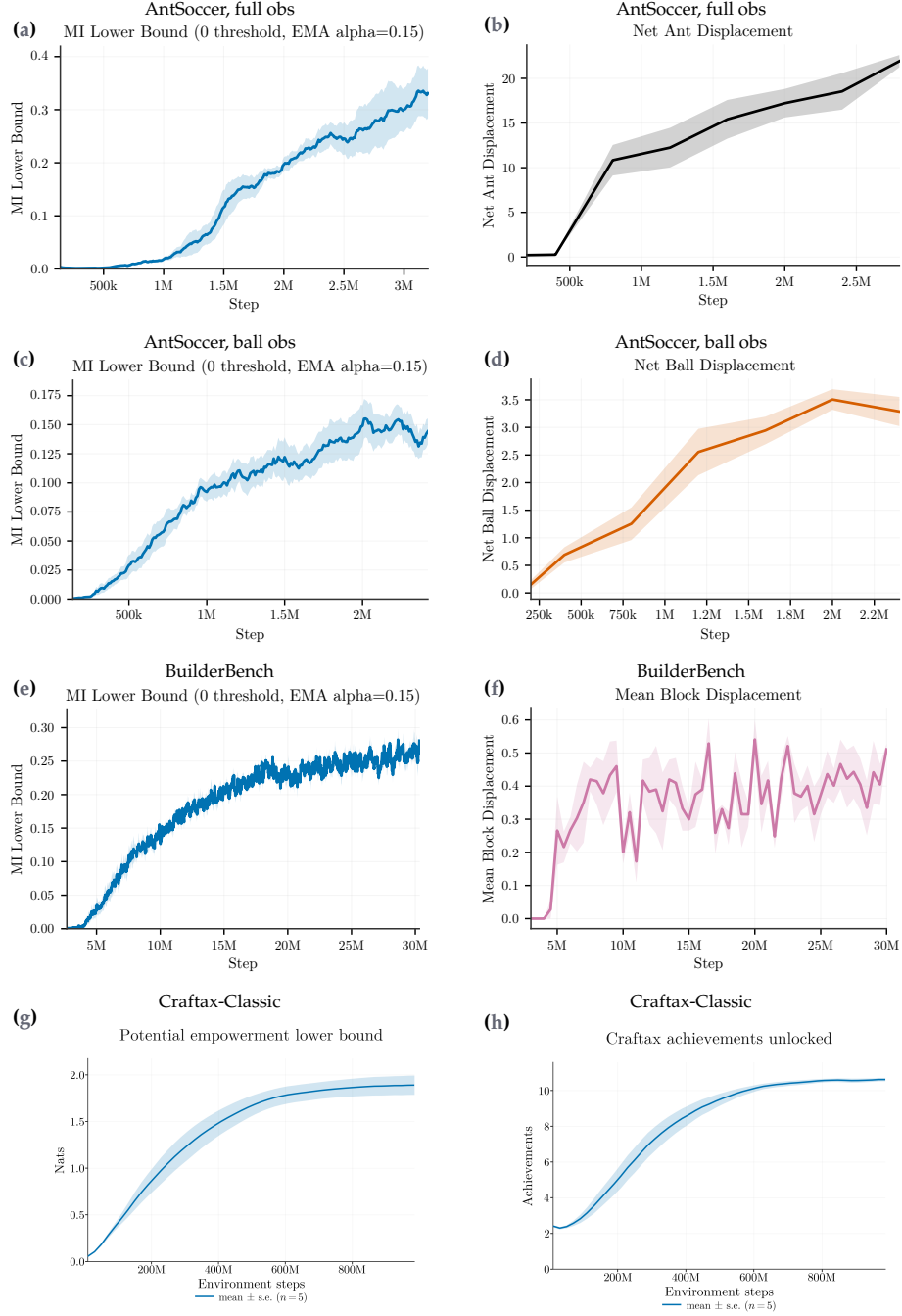


Figure 4: **The average potential empowerment estimate increases with control.** The estimated MI lower bound on potential empowerment increases over environment steps across all settings (left). Behavioral outcomes also improve: displacement of the controlled objects grows in the continuous settings (b, d, f), and the number of achievements unlocked grows in Craftax-Classic (h). The y-axis of the MI panels is in nats. The agent receives no external reward. (a, b): AntSoccer with full observations, over 5 seeds. (c, d): AntSoccer with observations restricted to the ball xy position, over 5 seeds. (e, f): BuilderBench with observations restricted to the cube xyz positions, over 4 seeds. (g, h): Craftax-Classic with the empowerment classifier restricted to the agent’s inventory, over 5 seeds.

6 Discussion

Despite the conceptual appeal of empowerment as a tool for understanding natural agents and building artificial agents, its seeming computational intractability has caused prior work to resort to approximations that do not learn one policy that “keeps its options open” [24].

In providing a conceptual link between skill learning methods and empowerment, we provide a method that maximizes potential and effective empowerment, two policies that measure controllability in distinct ways. We present key practical considerations to learn empowerment maximization policies in high-DoF settings. Finally, we present the results of empowerment maximization in a large environment, demonstrating that empowerment rewards control-seeking behavior distinct from that of common intrinsic reward methods. Future work should investigate simplified parameterizations of empowerment maximization that can resolve practical considerations detailed within our method, at scale. Furthermore, our work highlights the sensitivity of empowerment maximization to representations of the world, with open questions on the interplay between representations of the world and empowerment maximization.

References

- [1] Alison Gopnik. Empowerment as Causal Learning, Causal Learning as Empowerment: A Bridge Between Bayesian Causal Hypothesis Testing and Reinforcement Learning. 2024.
- [2] David Abel, Michael Bowling, André Barreto, Will Dabney, Shi Dong, Steven Hansen, Anna Harutyunyan, Khimya Khetarpal, Clare Lyle, Razvan Pascanu, et al. Plasticity as the Mirror of Empowerment. *Neural Information Processing Systems*, 38:164340–164374, 2026.
- [3] Khurram Javed and Richard S Sutton. The Big World Hypothesis and Its Ramifications for Artificial Intelligence. *Finding the Frame: An RLC Workshop for Examining Conceptual Frameworks*, 2024.
- [4] Karol Gregor, Danilo Jimenez Rezende, and Daan Wierstra. Variational Intrinsic Control. *International Conference on Learning Representations*, 2017.
- [5] Shakir Mohamed and Danilo Jimenez Rezende. Variational Information Maximisation for Intrinsically Motivated Reinforcement Learning. *Neural Information Processing Systems*, 28, 2015.
- [6] A.S. Klyubin, D. Polani, and C.L. Nehaniv. Empowerment: A Universal Agent-Centric Measure of Control. *IEEE Congress on Evolutionary Computation*, volume 1, pp. 128–135 Vol.1, 2005.
- [7] Tobias Jung, Daniel Polani, and Peter Stone. Empowerment for Continuous Agent—Environment Systems. *Adaptive Behavior*, 19(1):16–39, 2011.
- [8] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. Diversity Is All You Need: Learning Skills Without a Reward Function. *International Conference on Learning Representations*, 2018.
- [9] Seohong Park, Oleh Rybkin, and Sergey Levine. Metra: Scalable Unsupervised RL With Metric-Aware Abstraction. *International Conference on Learning Representations*, 2023.
- [10] Chongyi Zheng, Jens Tuyls, Joanne Peng, and Benjamin Eysenbach. Can a MISL Fly? Analysis and Ingredients for Mutual Information Skill Learning. *ICLR 2025*, 2025.
- [11] Michael Matthews, Michael Beukman, Benjamin Ellis, Mikayel Samvelyan, Matthew Jackson, Samuel Coward, and Jakob Foerster. Craftax: A Lightning-Fast Benchmark for Open-Ended Reinforcement Learning. arXiv:2402.16801, 2024.
- [12] Lucas Gruaz, Alireza Modirshanechi, Sophia Becker, and Johanni Brea. Merits of Curiosity: A Simulation Study. 2024.
- [13] Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell. Curiosity-Driven Exploration by Self-Supervised Prediction. *International Conference on Machine Learning*, pp. 2778–2787, 2017.
- [14] Marc Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, and Remi Munos. Unifying Count-Based Exploration and Intrinsic Motivation. *Neural Information Processing Systems*, 29, 2016.

- [15] Rein Houthoofd, Xi Chen, Yan Duan, John Schulman, Filip De Turck, and Pieter Abbeel. Vime: Variational Information Maximizing Exploration. *Neural Information Processing Systems*, 29, 2016.
- [16] Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by Random Network Distillation. *International Conference on Learning Representations*, 2019.
- [17] Philippe Capdepuy. Informational Principles of Perception-Action Loops and Collective Behaviours. 2011.
- [18] Eunice Yiu, Kelsey Allen, Shiry Ginosar, and Alison Gopnik. Empowerment Gain and Causal Model Construction: Children and Adults Are Sensitive to Controllability and Variability in Their Causal Interventions. arXiv:2512.08230, 2026.
- [19] Fei Dai and Alison Gopnik. Understand Intrinsic Motivation in Causal Learning Through Exploration. *Annual Meeting of the Cognitive Science Society*, volume 47, 2025.
- [20] Aly Lidayan, Yuqing Du, Eliza Kosoy, Maria Rufova, Pieter Abbeel, and Alison Gopnik. Intrinsically-Motivated Humans and Agents in Open-World Exploration. arXiv:2503.23631, 2025.
- [21] Claire Yang, Claire Jie Zhang, Maya Cakmak, and Max Kleiman-Weiner. When Assisting One Disempowers Another. arXiv:2511.04177, 2026.
- [22] Yuqing Du, Stas Tiomkin, Emre Kiciman, Daniel Polani, Pieter Abbeel, and Anca Dragan. Ave: Assistance via Empowerment. *Neural Information Processing Systems*, 33:4560–4571, 2020.
- [23] Andrew Levy, Alessandro G Allievi, and George Konidaris. Representation Learning and Skill Discovery With Empowerment. *Reinforcement Learning Conference*, 2025.
- [24] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. Keep Your Options Open: An Information-Based Driving Principle for Sensorimotor Systems. *PloS One*, 3(12):e4018, 2008.
- [25] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. All Else Being Equal Be Empowered. *European Conference on Artificial Life*, pp. 744–753, 2005.
- [26] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation Learning With Contrastive Predictive Coding. arXiv:1807.03748, 2018.
- [27] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning With a Stochastic Actor. *International Conference on Machine Learning*, pp. 1861–1870, 2018.
- [28] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight Experience Replay. arXiv:1707.01495, 2018.
- [29] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-Balanced Loss Based on Effective Number of Samples. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9260–9269, 2019.
- [30] Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep Variational Information Bottleneck. *International Conference on Learning Representations*, 2019.
- [31] Faisal Mohamed, Catherine Ji, Benjamin Eysenbach, and Glen Berseth. Temporal Representations for Exploration: Learning Complex Exploratory Behavior Without Extrinsic Rewards. 2026.
- [32] Raj Ghugare, Roger Creus Castanyer, Catherine Ji, Kathryn Wantlin, Jin Schofield, Karthik Narasimhan, and Benjamin Eysenbach. BuilderBench: The Building Blocks of Intelligent Agents. arXiv:2510.06288, 2026.
- [33] Seohong Park, Kevin Frans, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking Offline Goal-Conditioned RL. *International Conference on Learning Representations*, 2025.
- [34] Catherine Ji, Vivek Myers, Sergey Levine, and Benjamin Eysenbach. The Geometry of Empowerment. arXiv:TODO, 2026.
- [35] David Barber and Felix Agakov. The IM Algorithm: A Variational Approach to Information Maximization. *Neural Information Processing Systems*, volume 16, 2003.
- [36] Martin L Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2014.

A Algorithm and Implementation Details

Below, we include key algorithm and implementation details alongside ablations and conceptual explanations of design decisions. We emphasize that much of the implementation details and tricks apply not only to empowerment maximization (which requires scalable underlying MISL methods to learn skill *fields*), but also to unsupervised skill learning methods. Thu, we hope that the following explanations and experiments help advance the fields of both empowerment and unsupervised skill discovery.

A.1 Training the classifier: method details and considerations

Building on CSF: the classifier. In continuous settings, we build on Zheng et al. [10, CSF], a recent state-of-the-art skill-learning method, to lower-bound the potential empowerment $\mathcal{E}_{\text{pot}}(s) = \max_{\pi \in \Pi_{\mathcal{Z}}} I^{\pi}(Z; S_+ | S_0 = s_0)$ for a fixed starting state s_0 .

The original CSF objective estimates the one-step mutual information $I(Z; S, S')$, where $s \sim p^{\pi}(s)$ is the marginal DSOM and s' is the next state. Let $q_{\phi}(z) = \text{Unif}(\mathcal{Z}) = p(z)$ and fix the starting state. Set the one-step reward to the logit that estimates the log ratio:

$$r(s, a, s'; z) = f_{\phi}(s, s', z) \quad (17)$$

$$= \log \frac{q_{\phi}(z | s, s')}{p(z)} \quad (18)$$

$$\approx \log \frac{p(z | s, s')}{p(z)}. \quad (19)$$

Taking the expectation over the learned logit f_{ϕ} gives a variational lower bound on $I(Z; S, S')$ [10]:

$$\begin{aligned} \mathcal{O}_{\text{csf}}(\pi) &= \mathbb{E}_{p(z)} \left[\sum_t \gamma^t r(s_t, a_t, s_{t+1}; z) \right] \\ &= \mathbb{E}_{p^{\pi}(s, s', z)} \left[\log \frac{q_{\phi}(z | s, s')}{p(z)} \right] \\ &= \mathbb{E}_{p^{\pi}(s, s', z)} \left[\log \frac{p(z | s, s')}{p(z)} + \log \frac{q_{\phi}(z | s, s')}{p(z | s, s')} \right] \\ &= \mathbb{E}_{p^{\pi}(s, s', z)} \left[\log \frac{p(z | s, s')}{p(z)} \right] - \mathbb{E}_{p^{\pi}(s, s')} \mathbb{E}_{p^{\pi}(z | s, s')} \left[\log \frac{p(z | s, s')}{q_{\phi}(z | s, s')} \right] \\ &= \mathbb{E}_{p^{\pi}(s, s', z)} \left[\log \frac{p(z | s, s')}{p(z)} \right] - D_{KL}[p(z | s, s') \| q_{\phi}(z | s, s')] \\ &\leq I(Z; S, S') \end{aligned} \quad (20)$$

by positivity of the Kullback-Liebler divergence. Thus, training a classifier that distinguishes skill latent samples from $p(z | s, s')$ versus $p(z)$ provides a lower bound on the true MI.

Building on CSF: logit parameterization. A key property of both CSF and Metric-aware Abstraction (METRA) is the particular bilinear parameterization of the logit [9, 10]:

$$f_{\phi}(s, s', z) = z^{\top}(\phi(s') - \phi(s)).$$

This parameterization incentivizes the MISL algorithm to learn skills that spread representations $\phi(s)$ along linear degrees-of-freedom [9]. We refer the reader to [9] for more information on this parameterization, and why it is necessary to scale skill-learning methods to continuous locomotion tasks.

Estimating the log-ratio. Both CSF and METRA train skill policies conditioned on continuous skill spaces $\mathcal{Z}$. In CSF, the method learns the logit using the noise-contrastive objective InfoNCE [26]. Both of these methods learn representations ϕ that align the linear difference $\phi(s') - \phi(s)$ and the executed skill $z \in \mathcal{Z}$, while pushing apart the linear difference from other skills.

Unlike CSF, our presented results condition on a set of discrete skills. The reason for this is two-fold. First, as noted in [9], discrete skills *improved* the diversity of the downstream behavior. Second, it is much simpler to interpret the learned empowerment values with a discrete skill space. For example, suppose that the skill latent space is a hypersphere in latent space.

Following CSF, our method trains the log-ratio estimator with standard noise-contrastive objectives (e.g., InfoNCE) [26]. Importantly, CSF, like METRA, utilizes a linearized form for the logit:

$$\mathcal{O}_{\text{csf}}(\pi) \log \frac{q_\phi(z | s, s')}{p(z)} \leq I(Z; S, S'). \quad (21)$$

The conditional, long-horizon effective empowerment Definition 3.3 MI is a simple modification to the underlying classifier

$$\log \frac{q_\phi(z | s_0, s_+)}{q_\phi(z | s_0)}, \quad s_+ \sim p_z^\pi(s_+ | s_0), \quad z \sim p(z | s_0).$$

whose denominator $p(z | s_0)$ is generally intractable for high-dimensional z under standard contrastive estimation. This dependence turns out to be unnecessary: for continuous skill spaces, fixing $p(z | s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ and absorbing state-dependent skill selection into the policy loses no generality [34]. The uniform prior therefore removes the intractable $p(z | s_0)$ from the estimator entirely, so the long-horizon conditional MI becomes contrastively estimable. This is *not* generally true for discrete skill spaces, where a uniform prior over K skills restricts the source to the $1/K$ -quantized simplex [34].

Rather than predicting z from one-step transitions, we train a classifier predicting z from transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$. A sample of the one-step reward given (s_0, z, s') approximates

$$r_{\text{eff}}(s, a; z, s_0) \approx \mathbb{E}_{p(s' | s, a)} \left[\log \frac{q_\phi(z | s_0, s_+ = s')}{p(z)} \right], \quad (22)$$

which, in expectation over discounted timesteps and skills, gives $\mathcal{O}(\pi) \leq I^\pi(Z; S_+ | S_0 = s_0) = \mathcal{E}_{\text{eff}}(\pi)$. This is a small modification to CSF (Eq. (21)), yielding a practical method to maximize effective empowerment.

Telescoping properties of the bilinear classifier. The particular bilinear parameterization of the logit leads to interesting telescoping properties. For a fixed skill latent z , take the cumulative reward objective

$$\begin{aligned} \mathbb{E} \left[\sum_t r(s_t, a_t, s_{t+1}; z) \right] &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t z^\top (\phi(s_{t+1}) - \phi(s_t)) \right] \\ &= z^\top \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t (\phi(s_{t+1}) - \phi(s_t)) \right] \\ &= z^\top \mathbb{E} \left[\left(\sum_{t=0}^{\infty} \gamma^t (\phi(s_{t+1}) - \gamma \phi(s_{t+1})) \right) - \phi(s_0) \right] \\ &= z^\top \mathbb{E} \left[\left(\sum_{t=0}^{\infty} \gamma^t (1 - \gamma) \phi(s_{t+1}) \right) - \phi(s_0) \right] \\ &= \mathbb{E} \left[(1 - \gamma)^2 \mathbb{E}_{p_{\text{shift}}(s_+ | s_0, z)} [\phi(s_+)] - \phi(s_0) \right]. \end{aligned}$$

Note that the dependence of the cumulative reward on $\phi(s_0)$ should not change the policy gradient. Therefore, both of the following expressions are valid surrogate objectives:

$$\mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s_+)] \quad \text{or} \quad \mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s_+) - \phi(s_0)].$$

Thus, utilizing the bilinear parameterization, there is *no theoretical difference* between:

$$z^\top \mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s_+)] \quad \text{or} \quad z^\top \mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s_+) - \phi(s_0)] \quad \text{or} \quad z^\top \mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s'_+) - \phi(s_+)]$$

where $\phi(s'_+)$ simply refers to the next state.

That being said, there is a practical difference between these surrogate objectives. This effect is expected due to extensive results that show reward-shaping *does* lead to different training dynamics, shifting the balance between bias and variance despite theoretically leaving the overall policy gradient unchanged. We do not do extensive analysis of the variance of these parameterizations here.

The surrogate objective implies that a few lines of code can correctly modify the CSF classifier *assuming the bilinear logit form*. Rather than train the classifier with consecutive s and s' tuples, train the classifier with s and s_+ sampled from the shifted discounted state occupancy measure. The rollout and logit parameterization do not have to change.

Validity of the bilinear logit parameterization. While the bilinear logit provides potentially convenient telescoping properties, it remains unclear whether such a parameterization is valid. One possible shortcoming of this parameterization is that it implies some unnatural time symmetry properties, explicitly mentioned in [9]. Reversing the flow of time by sending $s \leftrightarrow s'$ and negating the selected skill does not change the logit value. Thus, this parameterization restricts the possible optimization space to the set of classifiers that respect this symmetry. While reasonable in a simple locomotion setting (e.g. equally easy to walk forward as it is to walk backwards), this property easily breaks in irreversible tasks like manipulation. For example, it is generally much harder to build a tower than it is to knock it down.

Another consequence of this bilinear logit is the assumption that skill-induced changes in the underlying state space should follow a potential and exhibit persistence over the $\frac{1}{1-\gamma}$ timescale. Otherwise, if there exists an uncontrollable probability flow sending $\phi(s')$ back to $\phi(s)$, a restricted-capacity skill space will naturally select other persistent degrees of freedom. Take, for instance, a simple setting of an ant in a bounded arena with an applied $+x$ -direction flow at some rate comparable to the $\frac{1}{1-\gamma}$ timescale. While, externally, we might consider swimming in the $+x$ -direction against the flow to be an equally valid skill as swimming in the $+/-y$ -direction, the $+x$ -direction cannot lead to persistent change. The telescoping property of the classifier shows that the distinguishability of a particular skill directly scales with the persistent perturbation

$$z^\top \mathbb{E}_{p_{\text{shift}}(s_+|s_0,z)}[\phi(s_+) - \phi(s_0)].$$

Thus, if the skill capacity is constrained, the policy will more readily choose the $+/-y$ -direction skill. In Craftax-Classic/Craftax (but not locomotion), we observe that this underlying assumption of the empowerment and logit parameterization prevents learning survival skills like ‘drink water’, ‘eat cow’, or ‘sleep’, which increases status bars that persistently decrease under environment dynamics. While these actions do lead to a large one-step increase in the distinguishability of the skill, the persistent effect of these behaviors are less than the effect of actions with persistence, such as collecting particular materials (unless the character actively consumes materials, materials do not disappear). Since the current methods to learn skills always, practically, have some underlying capacity constraint – if this were not the case, existing skill methods should learn much more diverse skills than those found in practice – the skills learned using the bilinear logit parameterization generally avoid non-persistent behaviors. Indeed, exchanging the bilinear logit parameterization with a fully-expressive skill classifier leads to skills that do successfully execute ‘eat cow’, ‘drink water’, and ‘wake up’ skills.

The above considerations suggest that learning more complex skill behaviors in tough manipulation settings or settings without persistent environment changes requires alternative formulations of the

logit parameterization and/or skill learning. While we believe this work to be very important for unsupervised skill discovery literature, developing entirely new skill learning methods extends beyond the scope of this paper which focuses on practical empowerment maximization.

A.2 Training value functions and policies

Representation stability. The theoretical statement implies that policies should train on

A.3 Accelerating skill-learning

On-policy implementations for GPU accelerated simulators. While many of the early RL benchmarks utilize CPU-based simulations, advances in the GPU stack have spawned new GPU-parallelized benchmarks that run $O(10^5)$ steps-per-second on a single modern GPU (e.g. all experiments in this paper ran on single L40 GPUs). Typically, methods that work best with large fresh environment-step-to-training-step ratios (e.g. on-policy methods like PPO) are particularly suited for accelerated simulator throughput.

Inconsistencies in MISL theory and practice. However, typical implementations of modern MISL methods utilize off-policy methods like SAC. Furthermore, most of these methods train (explicitly or implicitly) classifiers on replay buffers that mix occupancy statistics over many past iterations of the policy. As a side note, this implementation choice is particularly intriguing given that policy improvement requires perfectly-fit, on-policy classifiers. Indeed, we are not the first to point out this incongruence. [23] identifies and theoretically resolves this fundamental inconsistency by conditioning the classifier on policy parameters during training. Unfortunately, while providing valuable insight on this broken assumption, conditioning on policy parameters does not scale.

Here, we provide an implementation and a practical summary of a PPO-based CSF and empowerment maximization method that better suits GPU-parallelized simulators. While we do not purport to conceptually answer the theoretical and implementation inconsistencies in USD, we believe that empirically investigating this inconsistency can help provide valuable insights on MISL training dynamics.

B Statements and Proofs

Below, we include the key statement and proof that justifies the empowerment-maximization method. While the particular parameterizations and use of the underlying MISL methods in large settings is novel, we explicitly note that prior works such as VIC [4] have already shown versions of the same theoretical results. We include these statements for completeness and emphasize that the primary contribution of our work is an empirical demonstration of tractable empowerment maximization, rather than novel theoretical results.

B.1 Policy improvement of the potential empowerment lower-bound

Under the assumptions of a perfectly-fit classifier, we show policy improvement on a lower bound of the potential empowerment. More specifically, iteratively alternating (1) training a variational lower bound on on-policy skill distributions/classifier and (2) running policy improvement taking the trained classifier as a reward improves a lower bound on the potential empowerment.

The result below applies a standard variational lower bound on mutual information (the Barber-Agakov bound in Barber and Agakov [35]) which prior work Gregor et al. [4] applies to the effective empowerment maximization/skill-discovery problem. The proof additionally invokes the standard policy improvement statement for a fixed reward [36]:

Theorem B.1 (Policy improvement for the MI that measures empowerment). *Let $0 < \gamma < 1$ be the discount factor and let $\pi_n(a \mid s, z; s_0)$ be the current skill-conditioned policy at iteration n , with $z \sim p(z)$ sampled at s_0*

and held fixed. Fit an on-policy classifier $q_n(z \mid s_0, s_+)$ to $p^{\pi_n}(z \mid s_0, s_+)$ and define the $(1 - \gamma)$ -normalized action value

$$Q_{\text{eff},z}^{\pi_{\text{eff},n},q_n}(s, a; s_0) \triangleq (1 - \gamma) \mathbb{E}_{\pi_{\text{eff},n}(a \mid s, z; s_0)} \left[\sum_{k=0}^{\infty} \gamma^k \log \frac{q_n(z \mid s_0, s_{t+k+1})}{p(z)} \middle| s_t = s, a_t = a \right].$$

Let π_{n+1} be obtained by a greedy policy-improvement step with respect to $Q_z^{\pi_n}(s, a; s_0)$ for every z and s_0 . If the classifier is perfectly fit on-policy at each iteration, $q_n(z \mid s_0, s_+) = p^{\pi_n}(z \mid s_0, s_+)$, then

$$I^{\pi_{\text{eff},n+1}}(Z; S_+ \mid S_0) \geq I^{\pi_{\text{eff},n}}(Z; S_+ \mid S_0) \quad \forall n$$

which proves policy improvement under the assumptions of a perfectly-fit classifier.

Improving the estimated potential empowerment-seeking behavior reduces to standard RL with the same classifier intrinsic reward:

$$\widehat{\mathcal{E}}_{\text{pot}}^n(s) \approx \mathbb{E}_{p(z)} \left[V_{\text{eff},z}^{\pi_{\text{eff},n}}(s; s) \right] \approx I^{\pi_{\text{eff},n}}(Z; S_+ \mid S_0 = s) \quad (23)$$

$$\leq \mathcal{E}_{\text{pot}}^n(s) = \max_{\pi \in \Pi_Z} I^\pi(Z; S_+ \mid S_0 = s), \quad (24)$$

which, under the γ -discounted sum, lower-bounds the cumulative potential empowerment objective:

$$\mathcal{O}_p^n(\pi_{\text{pot}}) \leq \mathbb{E}_{\pi_{\text{pot}}} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}^n(s) \right].$$

The skill-conditioned effective empowerment policy $\pi_{\text{eff}}(a \mid s, z; s_0)$ and the potential policy $\pi_{\text{pot}}(a \mid s)$ are thus closely related but distinct.

Proposition B.2 (A fixed classifier defines an ordinary intrinsic-reward MDP). *Fix a classifier $q(z \mid s_0, s_+)$ and define the normalized intrinsic reward*

$$r_{\text{eff},q}(s_0, z, s') \triangleq (1 - \gamma) \log \frac{q(z \mid s_0, s')}{p(z)}.$$

For any policy π , define

$$Q_z^{\pi,q}(s, a; s_0) \triangleq \mathbb{E}_{\pi(a \mid s, z; s_0)} \left[\sum_{k \geq 0} \gamma^k r_q(s_0, z, s_{t+k+1}) \middle| s_t = s, a_t = a \right].$$

Then $Q_{\text{eff},z}^{\pi,q}$ satisfies the Bellman equation

$$\begin{aligned} Q_{\text{eff},z}^{\pi,q}(s, a; s_0) &= (1 - \gamma) \mathbb{E}_{p(s' \mid s, a)} \left[\log \frac{q(z \mid s_0, s')}{p(z)} \right] \\ &\quad + \gamma \mathbb{E}_{p(s' \mid s, a)} \left[V_{\text{eff},z}^{\pi,q}(s'; s_0) \right], \end{aligned}$$

where $V_{\text{eff},z}^{\pi,q}(s; s_0) = \mathbb{E}_{\pi(a \mid s, z; s_0)} Q_{\text{eff},z}^{\pi,q}(s, a; s_0)$.

Proof. Re-express the discounted sum in terms of the reward and remaining summation. The classifier is fixed, so $r_{\text{eff},q}$ is a stationary reward for the induced MDP indexed by (z, s_0) . $\square$

Proposition B.3 (Barber–Agakov lower bound is tight at the on-policy posterior). *For any policy π and any classifier $q(z \mid s_0, s_+)$,*

$$\begin{aligned} \mathcal{L}(\pi, q) &\triangleq \mathbb{E}_{p(z) p_z^\pi(s_+ \mid s_0)} \left[\log \frac{q(z \mid s_0, s_+)}{p(z)} \right] \\ &\leq I^\pi(Z; S_+ \mid S_0 = s_0). \end{aligned}$$

Equality holds if $q(z \mid s_0, s_+) = p^\pi(z \mid s_0, s_+)$.

Proof. Write the MI as

$$I^\pi(Z; S_+ | S_0 = s_0) = H(Z) + \mathbb{E}_{p^\pi(s_+ | s_0)} \mathbb{E}_{p^\pi(z | s_0, s_+)} [\log p^\pi(z | s_0, s_+)].$$

For every s_+ ,

$$D_{\text{KL}}(p^\pi(z | s_0, s_+) \| q(z | s_0, s_+)) \geq 0,$$

so replacing $\log p^\pi$ by $\log q$ gives a lower bound. Since $H(Z) = -\mathbb{E}_{p(z)} \log p(z)$ and the joint law factors as $p(z) p_z^\pi(s_+ | s_0)$, the bound is exactly $\mathcal{L}(\pi, q)$. Equality holds at the on-policy posterior. $\square$

Proof of Theorem B.1. At iteration n , the classifier q_n is fit to the on-policy posterior, so

$$\mathcal{L}(\pi_{\text{eff } n}, q_n) = I^{\pi_{\text{eff } n}}(Z; S_+ | S_0).$$

Now hold q_n fixed. By Proposition B.2, this is an ordinary MDP with stationary intrinsic reward. A greedy policy-improvement step therefore gives

$$\mathcal{L}(\pi_{\text{eff } n+1}, q_n) \geq \mathcal{L}(\pi_{\text{eff } n}, q_n).$$

Finally refit the classifier to the new on-policy posterior. By Proposition B.3, the refitted classifier makes the lower bound tight and can only increase the lower-bound value relative to the old classifier:

$$\begin{aligned} I^{\pi_{\text{eff } n+1}}(Z; S_+ | S_0) &= \mathcal{L}(\pi_{\text{eff } n+1}, q_{n+1}) \\ &\geq \mathcal{L}(\pi_{\text{eff } n+1}, q_n) \\ &\geq \mathcal{L}(\pi_{\text{eff } n}, q_n) \\ &= I^{\pi_{\text{eff } n}}(Z; S_+ | S_0). \end{aligned}$$

Thus exact on-policy classifier fitting plus greedy improvement monotonically improves the true MI. $\square$

B.2 Practical considerations to scale potential empowerment maximization

Drawing skills from the uniform distribution. As described in §. A.3, we roll out skills from a marginal and roll out un-conditioned uniform distribution $z \sim p(z)$ rather than fitting the conditional source $p(z | s_0)$. This contrasts with Variational Intrinsic Control [4], which assumes and fits arbitrary source distributions $p(z | s_0)$. Furthermore, VIC does not simultaneously learn a potential empowerment policy, that is, a policy that maximizes empowerment as defined in [6].

Necessity of Information Bottlenecks. In practice, maximizing the start state-conditioned MI in continuous settings leads to a failure mode: very small, almost imperceptible perturbations in state in, e.g., a deterministic settings can already saturate MI if using a constrained skill space. In fact, high MI *does not directly imply interesting behavior in continuous settings*.

Thus, it is critical to apply a form of information bottleneck from $S_0, S_+ \rightarrow Z$ when computing intrinsic rewards in the continuous settings. Adding Gaussian noise to states before feeding them into the classifier during training. This is a simplified case of applying the Variational Information Bottleneck [30]).

B.3 Pseudocode

The discrete and continuous skill settings share the same loop, with f_ϕ parametrized as in Eq. (11), and differ only in how f_ϕ is fit: a multi-class cross-entropy loss over the M skills when $\mathcal{Z}$ is discrete, and a contrastive loss (InfoNCE [26], following CSF [10]) when $\mathcal{Z}$ is continuous. In both cases the policy update uses f_ϕ directly as the reward.

Algorithm 2: MISL for empowerment maximization

Require: MDP $\mathcal{M}$, discount γ , skill space $\mathcal{Z}$, skill prior $p(z) = \text{Unif}(\mathcal{Z})$, rollout length M , iterations N , buffer capacities $|\mathcal{B}_q| \ll |\mathcal{B}_\pi|$

- 1: **Initialize:** $\pi_{\text{eff}}(a | s; z, s_0)$, $\pi_{\text{pot}}(a | s)$, $f_\phi(z; s_0, s_+)$, $\mathcal{B}_q \leftarrow \emptyset$ (small), $\mathcal{B}_\pi \leftarrow \emptyset$ (large)
- 2: **for** $n = 1, \dots, N$ **do**
- 3: **Rollout:** run π_{pot} for M steps, ending at handoff state s_0
- 4: **for** skills $z_k \stackrel{\text{i.i.d.}}{\sim} p(z)$, $k = 1, \dots, K_z$ **do**
- 5: run $\pi_{\text{eff}}(a | s, z_k, s_0)$ from s_0 for αM steps
- 6: sample s_+ from the discounted future states of this skill rollout
- 7: append (s_0, z_k, s_+) to $\mathcal{B}_q$ and $(s_t, a_t, s_{t+1}, s_0, z_k)$ to $\mathcal{B}_\pi$
- 8: **Classifier update:** fit $f_\phi(z; s_0, s_+) \approx \log \frac{p(z|s_0, s_+)}{p(z)}$ on $(s_0, z, s_+) \sim p_{\mathcal{B}_q}$: cross-entropy if $\mathcal{Z}$ discrete, contrastive if $\mathcal{Z}$ continuous
- 9: **Effective-policy update:** sample $s_+ \sim p_{\mathcal{B}_\pi}(s_+)$, HER sample the handoff state s_0 backwards in time from the same trajectory, and assign $z \leftarrow \arg \max_{z' \in \mathcal{Z}} f_\phi(z'; s_0, s_+)$
- 10: update π_{eff} on $Q_{\text{eff}, z}^{\pi_{\text{eff}}}(s, a; s_0)$ with reward $r_{\text{eff}, n} = f_\phi(z; s_0, s_+)$
- 11: **Potential-policy update:** update π_{pot} via SAC [27] on minibatches from $\mathcal{B}_\pi$, training $\hat{Q}_{\text{pot}}^{\pi_{\text{pot}}}$ with intrinsic reward $\widehat{\mathcal{E}}_{\text{pot}}(s) \approx \mathbb{E}_{p(z)}[V_{\text{eff}, z}^{\pi_{\text{eff}}}(s; s)]$
- 12: **return** $\pi_{\text{eff}}, \pi_{\text{pot}}$

B.4 Hyperparameters and training results

B.4.1 AntSoccer, full obs

Here empowerment is defined over every degree of freedom, the coordinates of both the ant and the ball:

$$\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; (S_{+, \text{ant}}, S_{+, \text{ball}}) | (S_{0, \text{ant}}, S_{0, \text{ball}})).$$

In this limit π_{pot} navigates to the center of the space and stays there (white, Fig. 3a), while the skills π_{eff} branch and cover the space (colors). The skills become more distinguishable because the agent first ascends the potential empowerment landscape. The estimated MI lower bound and the net agent displacement both grow over training, despite no external reward for locomotion.

Hyperparameters are in Table 1. Training curves are in Fig. 4a and Fig. 4b.

B.4.2 AntSoccer, ball obs only

Hyperparameters are in Table 2. Training curves are in Fig. 4c and Fig. 4d.

B.4.3 BuilderBench

Hyperparameters are in Table 3. The figures of MI over environment steps and displacement over environment steps are in Fig. 4e and Fig. 4f.

Table 1: Hyperparameters in AntSoccer, full obs

Parameters	Value
Environment	AntSoccer arena (navigate, single task)
Agent	potential empowerment
State dimension	42
Action dimension	8
Ball observations only	False
Center ball on reset	False
Agent start center fraction	0
Observation noise std (classifier bottleneck)	0.5
Number of skills K	30
Skill dimension d	4
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	5,000,000
Updates per env step	1
Replay buffer size	5,000,000
Skill buffer size	100,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (effective-number weighting [29])
Target entropy	0 (learned temperature, init 1)
Learning rate	3e-4
Batch size	256
Critic target update rate	0.02
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

Table 2: Hyperparameters in AntSoccer, ball obs only

Parameters	Value
Environment	AntSoccer arena (navigate, single task)
Agent	potential empowerment
State dimension	42
Action dimension	8
Ball observations only	True
Center ball on reset	True
Agent start center fraction	0.8
Observation noise std (classifier bottleneck)	0.05
Number of skills K	30
Skill dimension d	4
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	5,000,000
Updates per env step	1
Replay buffer size	5,000,000
Skill buffer size	100,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (effective-number weighting [29])
Target entropy	0 (learned temperature, init 1)
Learning rate	3e-4
Batch size	256
Critic target update rate	0.02
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

Table 3: Hyperparameters in BuilderBench

Parameters	Value
Environment	BuilderBench (3 cubes, corner cluster)
Agent	potential empowerment
State dimension	51
Action dimension	5
Cube observations only	True
Observation noise std (classifier bottleneck)	0.005 ($\approx 2.5\%$ of cube half-extent)
Number of skills K	30
Skill dimension d	9
Discrete skills	True
Potential policy steps M	100
Max episode steps	400
Online env steps	40,000,000
Env steps per update	8
Parallel environments	100
Replay buffer size	40,000,000
Skill buffer size	2,000,000
Value telescoping reference	fixed handoff state
Encoder reference	current state
Discrete reward softmax	True
Achieved-skill relabeling	True (argmax, effective-number weighting [29])
Target entropy	-12.5 (learned temperature, init 1)
Discount γ	0.99
Contrastive temperature	0.1
Encoder target update rate	0.99
Separate skill-0 actor and critic	True

C Discrete Experiment Implementation Details

C.1 Craftax-Classic

Hyperparameters are in Table 4. Fig. 5 shows one two-phase episode. We use PPO rather than SAC to utilize the GPU acceleration in the Craftax environment.

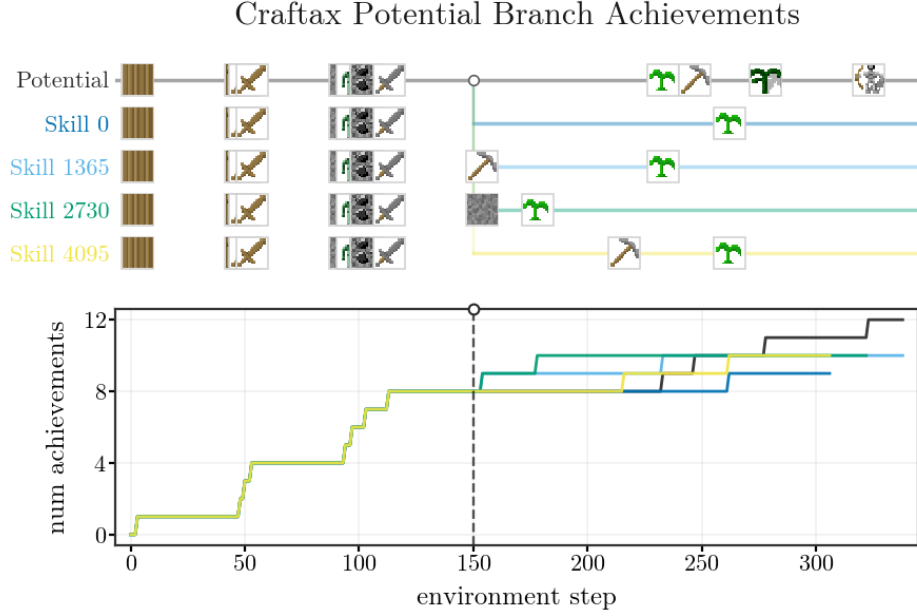


Figure 5: **Achievements in a single Craftax-Classic two-phase episode.** Icons mark achievement unlocks. Gray: shared potential phase up to the handoff at $T_{\text{pot}} = 150$ (dashed line). Colors: skill-conditioned continuations from the same handoff state. Bottom: cumulative achievements per rollout.

Table 4: Hyperparameters in Craftax-Classic

Parameters	Value
Environment	Craftax-Classic (symbolic)
Agent	potential empowerment (Potential-PPO)
Observation dimension	1345
Number of discrete actions	17
Parallel environments	8192
Encoder observations	inventory only
Observation noise std (classifier bottleneck)	0
Encoder norm cap c	5.0
Logit temperature τ	0.1 (fixed)
Encoder learning rate	3e-4
Encoder target update rate	0.99
Representation archive size	10,000,000
Encoder updates per policy update	4
Number of skills	16
Skill dimension	16 (standard-basis library)
Discrete skills	True
Potential policy steps	150
Max episode steps	1000
Online env steps	1,000,000,000
Reward	centered logit
Skill-0 reward	normalized potential estimate (Eq. (16))
GAE cut at phase switch	True
Separate skill-0 actor and critic	True
Learning rate	2.5e-4
Batch size	4096
Minibatches	4
Update epochs	4
Discount γ	0.99
GAE λ	0.95
PPO clip ϵ	0.2
Value loss coefficient	0.5
Entropy coefficient	0.01
Max gradient norm	0.5
Actor hidden dims	[512, 512, 512, 512]
Layer norm	True

Glossary

$\mathcal{E}_{\text{eff}}$ the effective empowerment as a function of the policy . 4, 6, 14, 19

$\mathcal{E}_{\text{pot}}$ The potential empowerment at state s and iteration n , defined as the channel capacity between skills and future states from that state. . 17

$\mathcal{E}_{\text{pot}}$ the potential empowerment as a function of the state . 3–6, 13, 17, 19, 24

π_{eff} an effective empowerment maximizing policy . 4–7, 17–19

π_{pot} a potential empowerment maximizing policy . 4–7, 17, 19

Acronyms

CSF contrastive successor features. 6, 13–16, 18

DSOM discounted state occupancy measure. 3, 13

MDP Markov decision process. 2, 3, 19

METRA Metric-aware Abstraction. 13, 14

MI mutual information. 2–4, 6–8, 13, 14, 16, 18, 19

MISL mutual information skill learning. 2–5, 13, 16, 19

RL reinforcement learning. 2, 17